Same Recipe, Different Fate: The Training Recipe's Effect on Safety Erosion from Benign Fine-Tuning Depends on the Model Family

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Abstract

Fine-tuning an aligned model on benign data can erode its safety, and published sizes of this erosion differ widely. We show that the effect of a training recipe on this erosion depends on the model family, and that which family it helps changes with model size. We fine-tune instruct models from three families (OLMo-2, Llama-3, Qwen2.5), at 1B and at 7–8B, with LoRA on 4,800 Dolly rows, three seeds per cell. The erosion E is the rise in the Unsafe share of 600 sampled answers to 200 harm prompts (Qwen3Guard), with HarmBench as a second judge. At 1B, a standard Scheduled recipe (warmup, cosine decay and a token-mean loss) cuts OLMo-2-1B's erosion from $+45.1$ to $+28.8$ pp but barely moves Llama-3.2-1B's ($+59.3$ to $+57.1$) or Qwen2.5-1.5B's ($+71.9$ to $+73.9$); both interactions exclude zero on both judges. In a pre-registered replication at 7–8B, the OLMo-versus-Llama interaction is not shown: OLMo-2-7B goes from $+44.1$ to $+50.3$ and Llama-3.1-8B from $+60.7$ to $+59.2$. There, Qwen2.5-7B is the family whose erosion falls ($+78.1$ to $+68.9$), on the main judge only. Halving the learning rate lowers erosion in both families alike. Under one recipe, OLMo erodes least and Qwen most at both sizes. The Unsafe label mostly tracks lost refusal (a blind human read). In a pre-registered test, dark fiction eroded no more than light fiction. We report every setting we screened and the judges' limits.

1 Introduction

Fine-tuning an aligned language model on benign data can erode its safety (Qi et al., 2024; He et al., 2024). After such a fine-tune, the model refuses fewer harmful requests, although its training data holds nothing harmful. Small open models are often fine-tuned on their users' own data. So it matters how large this erosion is, and what sets its size.

Published numbers differ widely. Each study uses its own model family, training recipe, data and judge. Family contrasts exist (Hawkins et al., 2026; Khan et al., 2026; Guo et al., 2026), and optimisation choices drive much of the safety loss (Kim et al., 2025). Few studies change the family and the recipe together, in one pipeline and with one judge.

This note is the measurement stage of a larger study on repairing safety after benign fine-tuning. A repair can only show its effect where the erosion is partial, so we first needed to know which families and recipes give partial erosion, and how stable it is across seeds. The repair study will appear separately, and nothing here depends on it.

We ask what sets the size of the erosion at about 1B and at 7–8B parameters. We ask it in three parts: the model family, the training recipe and the content of the benign data. We fine-tune six instruct models from three families (OLMo-2, Llama-3 and Qwen2.5, at both sizes) on the same 4,800 Dolly rows with one LoRA recipe, three seeds each. Then we change only the loss weighting and the learning-rate schedule, and test, with pre-registered runs, whether the recipe's effect carries across families and sizes. Last, in a pre-registered test, we replace Dolly with dark or light fiction of the same format and length. The erosion E is the rise in the share of sampled answers to harm prompts that a safety judge labels Unsafe.[1]

We make four contributions:

- **The recipe's effect depends on the family.** At 1B, a standard learning-rate schedule and loss cut OLMo's erosion by about a third, with over-refusal near the untuned level, but barely move Llama's or Qwen's. Both interactions exclude zero on two judges (§4, §5), and a loss-weighting-only recipe shows that

[1] All numbers come from development prompt sets; a held-out set is kept for the larger study (§2).

the schedule causes most of the cut.

- **The effect does not transfer across sizes.** In a pre-registered replication at 7–8B, the OLMo-versus-Llama interaction is not shown, and its point estimate has the opposite sign on both judges. At that size, Qwen is the family whose erosion falls, on the main judge only. Halving the learning rate lowers erosion in both families alike (§5).
- **The family sets the size at a fixed recipe.** One recipe gives partial erosion for OLMo-2 and near-saturation for Qwen2.5 and Llama-3, at 1B and at 7–8B (§3). We are not aware of earlier benign-fine-tuning erosion numbers for OLMo-2 Instruct (§10).
- **Content of benign data.** In a pre-registered test on OLMo-2-1B, dark fiction eroded no more than light fiction, and both eroded far less than Dolly at the same format and length, while refusing more benign prompts (§8).

We also locate the change. About half of OLMo's rise comes from answers that no longer open with a refusal (§6), and exploratory probes find that the schedule keeps more of OLMo-2-1B's refusal direction (§7). Section 9 lists every setting we screened, and the limits of the judges.

Our claims hold only for these models, this data, this prompt set and these judges. A rise in the Unsafe share alone does not show harmful compliance. We do not claim a cause for the family difference.

2 Setup

Models. We use six instruct models from three families, at two sizes: OLMo-2-0425-1B-Instruct and OLMo-2-1124-7B-Instruct (Team OLMo et al., 2024), Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct (Qwen et al., 2024), and Llama-3.2-1B-Instruct and Llama-3.1-8B-Instruct (Grattafiori et al., 2024). We write OLMo-2-1B, OLMo-2-7B, Llama-3.2-1B and Llama-3.1-8B. Section B gives the revision hashes. We call each model before our fine-tune the *untuned* model.

Data. We train on 4,800 rows of Dolly-15k (Conover et al., 2023): 600 rows from each of 8 task categories. Every run is one epoch, except the High-dose screen below. We tokenise the rows separately for each family, so the same rows give different token counts.

Recipes. Every recipe trains LoRA adapters (Hu et al., 2022) of rank 64 and alpha 128 on all linear layers, with AdamW. Training runs in our own PyTorch loop, with Hugging Face Transformers and PEFT, and Table 6 lists every setting. The *Constant* recipe uses a constant learning rate of 10^{-4}, no warmup, 128 rows per step and one epoch (38 steps). Its loss is a per-row mean: each row weighs the same, so the tokens of short answers weigh more. The *Scheduled* recipe is the Constant recipe plus a 4-step linear warmup with cosine decay from a peak learning rate of 10^{-4}, and a token-mean loss, so each response token weighs the same. The Constant recipe is not standard practice; the Scheduled recipe is the standard-practice comparison. Each variant changes one thing: *Constant, token-mean* changes only the loss weighting; *Constant, half LR* and *Scheduled, half LR* halve the (peak) learning rate to 5×10^{-5}; *Constant, rank 16* and *Scheduled, rank 16* use rank 16 and alpha 32. The 7–8B models use the same recipes with 2 rows per micro-batch and 64 accumulation steps, so the same 128 rows per step. On Qwen2.5-1.5B only, we first screened a *High-dose* recipe: a constant learning rate of 2×10^{-4}, 8 rows per step (600 steps per epoch) and 3 epochs.

Evaluation. The 200 development harm prompts come from HarmBench's standard behaviours (Mazeika et al., 2024), StrongREJECT (Souly et al., 2024), SimpleSafetyTests (Vidgen et al., 2024) and the toxic set of OR-Bench (Cui et al., 2025). For each prompt, we sample 3 answers at temperature 0.7 (top-p 1.0), and we take one greedy answer. Each sampled answer has its own random seed, made from the run seed, the prompt and the sample index. An answer may have up to 2,048 new tokens. Over-refusal uses 100 benign prompts: 50 safe prompts from XSTest (Röttger et al., 2024) and 50 from OR-Bench-Hard (Cui et al., 2025). It is the share of the validly judged greedy answers to these prompts that Qwen3Guard (below) labels as a refusal (*Refusal: Yes*). Task skill is the held-out loss on 200 Dolly rows. We also report ARC forced choice on 189 questions (Clark et al., 2018).

Judges and erosion. The main judge is Qwen3Guard-Gen-4B (Zhao et al., 2025a). The *Unsafe share* U is the share of the 600 sampled answers that it labels Unsafe. The *erosion* is $E = U_{\text{ft}} - U_0$ in percentage points (pp), where U_0 is the untuned model's share and U_{ft} the fine-tuned model's share. U is a broad policy label, not

an attack success rate (§9). Granite Guardian 3.3 8B (Padhi et al., 2024), in its 4-bit Q4_K_M build with its harm criterion, is a second judge on the 200 greedy answers. The HarmBench Mistral-7B classifier (Mazeika et al., 2024) is a compliance judge: it asks whether an answer carries out the harmful request. As in HarmBench's code, it reads at most the first 511 tokens of an answer. We write its erosion as E_{HB}. Section B also gives the judges' revisions.

Statistics. We cluster the standard error (SE) of U by prompt, because the 3 answers to one prompt are correlated; the design effect is about 2. The band for E is $E \pm 2\,\mathrm{SE}$. It covers evaluation noise only. With 3 seeds, the seed standard deviation (SD) has 2 degrees of freedom (df).

Development prompts only. Every number in this note comes from the development prompt sets above; the larger study keeps a separate held-out set, which we did not use. So no result here is confirmed on prompts outside the development set, and some choices, such as the partial window (§3), were made after seeing development results (§9).

Release. We will release the recipe files, the per-answer judge labels (no prompt or answer text), the analysis scripts and the figure data with the final version.

3 Same recipe, different fate

Table 1 gives the erosion under the Constant recipe, three seeds per model. OLMo-2-1B rises from 3.2% to a mean of 48.3%: the mean E is $+45.1$ pp (band 41.3–49.0). Llama-3.2-1B rises by $+59.3$ pp and Qwen2.5-1.5B by $+71.9$ pp. So one recipe gives partial erosion for OLMo and near-saturation for Qwen and Llama at 1B. The 7–8B models keep this order: OLMo-2-7B rises by $+44.1$ pp, Llama-3.1-8B by $+60.7$ pp and Qwen2.5-7B by $+78.1$ pp. The seeds spread more at 7–8B (sample SDs of 6–8 pp, against 3–5 pp at 1B). We call erosion *partial* if E lies between 20 and 55 pp, the range that the larger study needs (§1). We fixed this window after the first Qwen results (§9). Both OLMo means lie inside it; the Llama and Qwen means lie above it. Figure 2 shows every seed of the 1B models.

The starting points differ: Llama-3.2-1B starts at 13.8%. So we also report the share of the headroom that each model uses, $E/(100 - U_0)$ (Table 1); the order of the models does not change. Over-refusal falls in every family. Qwen2.5-1.5B starts at 68% and Qwen2.5-7B at 7%, and both near saturation, so a high starting over-refusal does not by itself explain Qwen's near-saturation. Held-out loss falls in every run, so each model learns the task. The compliance judge keeps the order of the families: E_{HB} is $+30.3$ pp for OLMo-2-1B, $+37.9$ for Llama-3.2-1B and $+45.4$ for Qwen2.5-1.5B at 1B, and $+33.1$, $+44.4$ and $+60.6$ pp at 7–8B (Table 1). It sees 0.63–0.78 of the Qwen3Guard erosion in every model, so most of the lost refusal is also harmful compliance as HarmBench defines it (§9).

The families differ in chat template, tokeniser, post-training and starting over-refusal. We do not claim a cause for the family difference.

4 The recipe matters

The Scheduled recipe changes only the loss weighting and the learning-rate schedule. On OLMo, it cuts the mean erosion from $+45.1$ to $+28.8$ pp, with 3 seeds of each recipe (Table 2). On seed 1101, whose two runs share their evaluation sampling streams, the difference is -14.5 pp, about 5 times the between-seed SD under the Constant recipe (2.65 pp, 2 df). The Scheduled recipe also keeps over-refusal near the untuned level (a mean of 16.3%, against 17% untuned and 4.7% under the Constant recipe), and its held-out loss is a little higher (1.613 against 1.545). Its three-seed band is 25.3–32.4 pp ($E \pm 2$ SE, evaluation noise only), and Granite agrees in direction on every seed (mean greedy E $+22.3$ against $+43.0$ pp). On the HarmBench compliance judge, with every OLMo unit judged on one GPU type, the mean E_{HB} is $+30.3$ pp under the Constant recipe and $+17.2$ pp under the Scheduled recipe: the Scheduled recipe keeps 0.57 of the Constant recipe's erosion on this judge, against 0.64 on Qwen3Guard. Under Scheduled, rank 16, the erosion is $+20.2$ pp on the same seed (8.8 pp lower); by a rule fixed before the run, rank matters little, but one seed cannot exclude a real difference of this size.

The Scheduled recipe makes two changes at once. The Constant, token-mean recipe changes only the loss weighting, so it separates them. Over three seeds it gives $E = +41.9$ pp ($+43.7$, $+39.5$ and $+42.5$), against $+45.1$ pp under the Constant recipe and $+28.8$ pp under the Scheduled recipe. So, on the main measure, the loss weighting accounts for about a fifth of the cut (3.2 of 16.3 pp; the Welch SE of that difference is 2.0 pp, so its share is not distinguishable from zero), and the schedule

Table 1: Same recipe, different fate. Every row uses the Constant recipe for one epoch, three seeds per model; Section A gives every seed. Values are means over the seeds. U_0 and U_{ft} are the Unsafe shares before and after fine-tuning. $\pm$ is the sample SD of E across seeds.

Model	Seeds	U_0	U_{ft}	E (pp)[a]	E_{HB} (pp)[b]	Over-refusal, % (0 → ft)	Held-out loss (0 → ft)
OLMo-2-1B	3	3.2%	48.3%	$+\mathbf{45.1} \pm 2.6$	+30.3	17 → 4.7	2.100 → 1.545
OLMo-2-7B	3[e]	1.8%	45.9%	$+\mathbf{44.1} \pm 6.4$	+33.1	27 → 2.3	2.121 → 1.355
Qwen2.5-1.5B	3[e]	4.7%	76.6%	$+\mathbf{71.9} \pm 3.0$	+45.4	68[c] → 2.3	1.890 → 1.462
Qwen2.5-7B	3[e]	4.8%	82.9%	$+\mathbf{78.1} \pm 4.4$	+60.6	7 → 2.7	3.335 → 1.295[d]
Llama-3.2-1B	3	13.8%	73.1%	$+\mathbf{59.3} \pm 4.6$	+37.9	11 → 3.0	2.243 → 1.554
Llama-3.1-8B	3[e]	5.3%	66.1%	$+\mathbf{60.7} \pm 8.2$	+44.4	17 → 1.0	2.109 → 1.323

[a]Headroom used, $E/(100 - U_0)$: OLMo-2-1B 0.47, OLMo-2-7B 0.45, Qwen2.5-1.5B 0.75, Qwen2.5-7B 0.82, Llama-3.2-1B 0.69, Llama-3.1-8B 0.64. [b]HarmBench on A100 pods, each fine-tuned unit against its own model's untuned unit (untuned shares: OLMo-2-1B 3.2%, OLMo-2-7B 1.2%, Qwen2.5-1.5B 3.2%, Qwen2.5-7B 5.3%, Llama-3.2-1B 11.0%, Llama-3.1-8B 5.5%). [c]The 68% untuned over-refusal is Qwen2.5-1.5B only; Qwen2.5-7B starts at 7% and still nears saturation. [d]The untuned 7B held-out loss is above the 1.5B's on the same rows, so we do not compare task loss across sizes. [e]Run on a later engine identity than the OLMo-2-1B and Llama-3.2-1B rows (Section B); seed 1101 of each Qwen model reproduced the earlier one-seed unit's counts exactly.

for the rest: a 4-step warmup and a cosine decay, which also halves the mean learning rate over the epoch. HarmBench gives the same split: E_{HB} is +27.7 pp under Constant, token-mean (three seeds), so the loss weighting accounts for 2.6 of the 13.2 pp cut on this judge too. We had predicted the opposite before the first token-mean run ($E \leqslant 36.25$ pp, pre-registered); the three-seed mean also fails that prediction. On the 200 greedy answers the loss weighting matters more (greedy E +33.3 against +41.5 pp; Granite +34.7 against +43.0 pp; three seeds each), and the opening lexicon agrees with the main measure (§6): the schedule keeps refusal openings, and the loss weighting mainly changes answer length. This agrees with two earlier findings: optimisation choices drive much of the safety loss (Kim et al., 2025), and a warmup alone cut erosion on benign data (Qi et al., 2025).

On Llama-3.2-1B, the Scheduled recipe barely moves the erosion: a mean of +57.1 pp over three seeds (SE 2.55), against +59.3 pp under the Constant recipe. Granite agrees in direction on every seed. So the Scheduled recipe cuts OLMo's erosion more than Llama's: the change is −16.3 pp for OLMo (−0.69 on the logit scale) and −2.2 pp for Llama (−0.11). By the fixed test below, the recipe effect differs between OLMo and Llama. So the family contrast survives the standard recipe: OLMo stays partial and Llama stays near saturation. The compliance judge shows the same pattern: the Scheduled recipe keeps 0.57 of OLMo's E_{HB} (+30.3 → +17.2 pp) but 0.94 of Llama's (+37.9 → +35.8 pp; three seeds each). Scheduled, half LR (a peak learning rate of 5×10^{-5}) moves OLMo from +28.8 to +20.8 pp (two seeds) and Llama from +57.1 to +41.2 pp (one seed): about −28% in both families (−0.40 and −0.69 on the logit scale). So the learning rate moves both families, while the Scheduled recipe moves OLMo far more than Llama. HarmBench did not judge the 1B half-LR runs; §5 repeats the comparison at 7–8B, pre-registered.

The logit scale and our claim rule. Near saturation, a fixed effect on the logit scale gives a smaller change in percentage points, so we compare recipes on both scales. The test takes the logit of U for each seed. For each family, it takes the mean under the Scheduled recipe minus the mean under the Constant recipe, with a variance from the seeds. The interaction is OLMo's change minus Llama's, with a Welch interval. We say that the recipe effect differs between the families only if the 95% interval excludes 0; otherwise we describe a pattern. The interaction is −0.57 (SE 0.17, Welch df 5.1; 95% interval −1.02 to −0.13). It excludes 0, so the Scheduled recipe cuts OLMo's erosion more than Llama's. In percentage points, the contrast is −14.1 pp (SE 3.7), and its interval also excludes 0. Limits go with this result. We wrote the exact test after 11 of the 12 values were known; the design (3 seeds of each recipe in both families, on the logit scale) was in our draft before Llama's Constant seeds 1102 and 1103 ran. The interval rests on about 5 df. Its judge is Qwen3Guard. The same test on the HarmBench labels, run once HarmBench had judged all twelve units, gives −0.60 (SE 0.14, Welch df 5.6; 95% interval −0.95 to −0.25), or −11.0 pp (−18.3 to −3.7), so the second judge

Table 2: The recipe matters. One line per model and recipe, with means over the seeds; Section A gives every seed. $\pm$ is the sample SD of E across seeds (3-seed rows only). Over-refusal is in %. Untuned values are in Table 1; §2 defines the recipes. The rows below the second rule are the pre-registered cells of §5.

Model	Recipe	Seeds	E (pp)	E_{HB} (pp)[a]	Over-refusal	Held-out loss	Granite E (pp)[b]
OLMo-2-1B	Constant	3	$+45.1 \pm 2.6$	$+30.3$	4.7	1.545	$+43.0$
OLMo-2-1B	Constant, token-mean	3	$+41.9 \pm 2.2$	$+27.7$	7.3	1.589	$+34.7$[c]
OLMo-2-1B	Scheduled	3	$+\mathbf{28.8} \pm 2.3$	$+17.2$	**16.3**	1.613	$+22.3$
OLMo-2-1B	Scheduled, half LR	2	$+20.8$	—	20.0	1.635	$+11.8$
OLMo-2-1B	Scheduled, rank 16	1	$+20.2$	—	25	1.649	$+9.0$
Llama-3.2-1B	Constant	3	$+59.3 \pm 4.6$	$+37.9$	3.0	1.554	$+62.8$[c]
Llama-3.2-1B	Scheduled	3	$+57.1 \pm 2.6$	$+35.8$	7.3	1.615	$+56.7$
Llama-3.2-1B	Scheduled, half LR	1	$+41.2$	—	4	1.642	$+38.0$
Qwen2.5-1.5B	Constant	3	$+71.9 \pm 3.0$	$+45.4$	2.3	1.462	$+80.2$[c]
Qwen2.5-1.5B	Scheduled	3	$+73.9 \pm 0.7$	$+52.7$	6.7	1.512	$+81.8$[c]
OLMo-2-7B	Constant	3	$+44.1 \pm 6.4$	$+33.1$	2.3	1.355	$+44.8$[c]
OLMo-2-7B	Scheduled	3	$+50.3 \pm 8.1$	$+37.8$	6.3	1.411	$+47.2$[c]
OLMo-2-7B	Constant, half LR	3	$+\mathbf{31.9} \pm 1.4$	$+24.0$	5.3	1.370	$+35.0$[c]
Llama-3.1-8B	Constant	3	$+60.7 \pm 8.2$	$+44.4$	1.0	1.323	$+59.3$[c]
Llama-3.1-8B	Scheduled	3	$+59.2 \pm 3.3$	$+42.5$	5.3	1.364	$+58.5$[c]
Llama-3.1-8B	Constant, half LR	2	$+53.3$	$+37.0$	2.0	1.327	$+52.5$[c]
Qwen2.5-7B	Constant	3	$+78.1 \pm 4.4$	$+60.6$	2.7	1.295	$+84.7$[c]
Qwen2.5-7B	Scheduled	3	$+\mathbf{68.9} \pm 1.9$	$+55.6$	5.7	1.336	$+73.7$[c]

—: HarmBench did not judge these runs. [a]HarmBench on A100 pods, every seed, each unit against its model's untuned unit (shares in Table 1); units judged in two pod runs got the same labels. [b]Granite on the 200 greedy answers, against the untuned unit judged on the same device; [c]on the pod (Mac-versus-pod parity passed; untuned pod shares in Section A). On the Mac, Llama-3.2-1B's seed 1101 gives $+67.5$.

agrees; Granite agrees in direction on every seed. §5 applies the same test, pre-registered, to a third family and to the 7–8B models.

5 Does the recipe effect transfer across sizes and families?

Does the OLMo-versus-Llama interaction hold at a larger size, and does a third family, Qwen, behave like OLMo or like Llama? We answered both with pre-registered runs. Before any of them, we fixed the cells, the test, the labels and our predictions in four files, with the test script hashed and self-tested on the 1B counts (Section B): (i) a 7–8B replication of the OLMo-versus-Llama test, 3 seeds per cell; (ii) the Scheduled recipe on Qwen2.5-1.5B, with T1 (OLMo-1B against Qwen-1.5B) and T2 (Qwen-1.5B against Llama-1B); (iii) the same on Qwen2.5-7B, with T5 (OLMo-7B against Qwen-7B) and T6 (Qwen-7B against Llama-8B); and (iv) the Constant recipe at half its learning rate on OLMo-2-7B (3 seeds) against Llama-3.1-8B (2 seeds). Every test is the test of §4 (Table 3), also run on the HarmBench labels. Six tests ran on overlapping cells with no multiplicity adjustment, and every unit is reported. Table 2 gives the cells.

At 1B, the Scheduled recipe lowers erosion only in OLMo. Qwen2.5-1.5B goes from $+71.9$ to $+73.9$ pp under the Scheduled recipe ($D = +0.11$). T1 reads "differs: OLMo's change is more negative" on both judges. T2 is "not shown" on Qwen3Guard but "differs" on HarmBench, where Qwen's E_{HB} rises from $+45.4$ to $+52.7$ pp while Llama's falls by 2 pp: so Qwen-1.5B's small rise is judge-dependent. We had predicted that Qwen behaves like Llama (D between -0.4 and 0; mean E between $+55$ and $+75$ pp), T1 "differs" and T2 "not shown": the mean is inside the range, D is just above it, and both test predictions held.

At 7–8B, the interaction does not replicate. Under the Scheduled recipe, OLMo-2-7B's mean E rises from $+44.1$ to $+50.3$ pp ($D = +0.25$) and Llama-3.1-8B's falls from $+60.7$ to $+59.2$ (-0.08). The interaction is "not shown" on both judges, with the opposite sign to 1B. We had predicted that it replicates and that OLMo-2-7B's Scheduled mean is about $+29$ pp (range $+15$ to $+40$): both failed; the Llama prediction (about $+65$, range $+55$ to $+75$) held. Granite agrees in direction with Qwen3Guard on every cell mean. The seeds spread more at 7–8B, so this test has less power than the 1B test, and "not shown" is weak evidence of no interaction. The sign is the point: the schedule did not cut OLMo-2-7B's erosion at all.

Table 3: The pre-registered tests, and the 1B test of §4. D is a family's mean logit U under the changed recipe minus under the Constant recipe (Qwen3Guard); $I = D_{\text{first}} - D_{\text{second}}$ with its Welch 95% interval, on Qwen3Guard and on the HarmBench labels (whose own D differ). "Differs" names the family whose change is more negative when the interval excludes 0; otherwise "not shown".

Test	First − second	D_{first}	D_{second}	I [95%], Qwen3Guard	I [95%], HarmBench	Label
1B test (§4; written late)	OLMo-1B − Llama-1B	−0.69	−0.11	−0.57 [−1.02, −0.13]	−0.60 [−0.95, −0.25]	differs (OLMo)
T1	OLMo-1B − Qwen-1.5B	−0.69	+0.11	−0.80 [−1.13, −0.46]	−0.98 [−1.30, −0.65]	differs (OLMo)
T2	Qwen-1.5B − Llama-1B	+0.11	−0.11	+0.22 [−0.24, +0.69]	+0.38 [+0.08, +0.67]	not shown; HarmBench differs
Replication, 7–8B	OLMo-7B − Llama-8B	+0.25	−0.08	+0.33 [−0.49, +1.16]	+0.28 [−0.39, +0.96]	not shown
T5	OLMo-7B − Qwen-7B	+0.25	−0.57	+0.82 [+0.08, +1.56]	+0.42 [−0.09, +0.93]	differs (Qwen); HarmBench not shown
T6	Qwen-7B − Llama-8B	−0.57	−0.08	−0.48 [−1.27, +0.30]	−0.14 [−0.80, +0.53]	not shown
Half LR, 7–8B	OLMo-7B − Llama-8B	−0.51	−0.34	−0.17 [−0.95, +0.60]	−0.13 [−0.79, +0.53]	not shown

At 7B, Qwen is the family that responds, on the main judge. Qwen2.5-7B's erosion falls from +78.1 to +68.9 pp under the Scheduled recipe ($D = -0.57$), the largest change of any 7–8B cell, with every Scheduled seed below every Constant seed. T5 reads "differs: Qwen's change is more negative" on Qwen3Guard, but its interval clears 0 only narrowly (+0.08), and on HarmBench it is "not shown" (E_{HB} falls from +60.6 to +55.6 pp). T6 is "not shown" on both judges. We had predicted that Qwen-7B barely moves (D between −0.5 and +0.2) and that T5 and T6 are "not shown": the mean is inside the predicted range (+65 to +85) but D is below it, and the T5 prediction failed. So Qwen-7B's response is clear on the main judge and smaller on the compliance judge.

The learning rate moves both families. Halving the Constant recipe's learning rate lowers OLMo-2-7B's mean E from +44.1 to +31.9 pp (3 seeds; $D = -0.51$) and Llama-3.1-8B's from +60.7 to +53.3 (2 seeds; −0.34). The interaction is "not shown" on both judges, as predicted, and OLMo-2-7B's half-LR mean (predicted about +32, range +20 to +45) held. The idea for this test came from interim replication data, as its file discloses. As at 1B, the learning-rate level moves both families, while the schedule change moves them differently.

Reading. Which family responds to the same recipe change depends on the size: at 1B OLMo, at 7–8B Qwen (on the main judge), and never both. So the effect of this standard recipe change transfers neither across families at one size nor across sizes within a family. Robust on both judges: the 1B OLMo effect, the non-replication at 7–8B and the learning-rate result; judge-dependent: Qwen-7B's response (T5) and Qwen-1.5B's small rise (T2).

6 Where the change sits

We sort each answer by its opening words into a refusal, a caveat or a direct opening, with a frozen word list (Section B). Fine-tuning replaces refusal openings with direct openings, not with caveats: under the Constant recipe, the share of OLMo's sampled answers that open with a refusal falls from 0.80 to 0.30 (to 0.44 under the Scheduled recipe), and Qwen2.5-1.5B's from 0.93 to 0.07 (Figure 1). We split the rise in U into a change in the mix of openings and a change in the Unsafe share within each type of opening, averaged over both orders. The opening switch carries about half of OLMo's rise (0.42–0.64 across word lists and judges; 0.48 under the Scheduled recipe) and more of Qwen's (0.64–0.78). Answers that open directly also became less safe: OLMo's Unsafe share within direct openings rose from 12 of 66 answers to 840 of 1,197 (pooled over 3 seeds), so the first words are not the whole account (compare the shallow-alignment view of Qi et al. (2025)). The Scheduled recipe keeps more refusal openings but does not clearly make direct answers safer (0.60 against 0.68 Unsafe within them, seed 1101).

The families differ in two layers, and the recipes

act through the openings; Section E gives the details and Figure 1 the opening mix.

7 Exploratory: where the recipe acts

Refusal direction. These analyses were run after the results above, carry no test and rest on one to three seeds per cell: leads for the larger study, not claims. For 13 fine-tuned units we take the residual stream after each decoder block at the last prompt token, for the 200 harm prompts and 200 harmless task prompts, in the untuned and the fine-tuned model. At each layer, the refusal direction r is the difference of the two groups' untuned means, and the separation is the distance between the groups' mean projections on r in pooled-SD units. Table 10 gives the share of the untuned separation that the fine-tuned model keeps and the cosine between the untuned and fine-tuned directions, averaged over the upper half of the layers. The Scheduled recipe keeps far more of OLMo-2-1B's separation (0.71–0.74 against 0.59; cosine 0.79–0.80 against 0.67), the one cell where it cuts erosion most, and little more of Llama-3.2-1B's (0.62 against 0.57) or OLMo-2-7B's (0.42 against 0.37–0.41). Llama-3.1-8B breaks the pattern: the recipe keeps more of its direction (0.41 against 0.32–0.35) without lowering its erosion. Every 7–8B unit keeps less than any 1B unit.

Other leads. Answer length, over-refusal, which prompts erode and the untuned starting points are in Section E.

8 Does the content of benign data matter?

We tested the third part with a pre-registered experiment on OLMo-2-1B under the Constant recipe (Section D). Two arms replace Dolly with passages from public-domain novels, in Dolly's format and length distribution: *dark* fiction (Gothic and horror books; the passages richest in a darkness lexicon) and *light* fiction (romance and domestic books; the passages poorest in it), each screened by Granite and run on the three Dolly seeds. We fixed the contrasts, margins and labels before any run.

Content did not raise the erosion. Dark minus light is -4.9 pp on E (95% interval -9.5 to -0.3) and -1.7 pp on E_{HB} (-5.1 to $+1.8$). Both 90% intervals lie inside the pre-registered margins (± 15.8 and ± 10.6 pp, about a third of Dolly's erosion on each judge), so the label is "no difference larger than the margin" (Table 8). The margins are wide, so this is a weak equivalence claim: content adds less than about a third of Dolly's erosion. We had predicted that dark fiction erodes more; the point estimates have the opposite sign, and on E alone the 95% interval excludes 0 in that direction (on E_{HB} it does not).

Fiction eroded far less than Dolly. Light fiction minus Dolly is -28.5 pp on E (-33.4 to -23.6) and -23.6 pp on E_{HB} (-28.4 to -18.8), so the label is "fiction differs from Dolly (lower)". Granite agrees in direction. E_{HB} is below 20 pp in both fiction arms, so their erosion is lost refusal, not harm (§9).

The fiction models also changed in other ways. They refused more benign prompts and lost ARC accuracy, so weaker instruction following may explain part of the smaller erosion (Section D).

9 What the measure means

What U tracks. U is a broad policy label. After fine-tuning, the probability of an Unsafe label given no refusal is 0.91–0.94, so U mostly tracks lost refusal; under hard categories only, OLMo's E is about 28 pp. HarmBench sees about two thirds of OLMo's Qwen3Guard erosion under the Constant recipe ($\kappa = 0.65$–0.71 with Qwen3Guard on the fine-tuned runs).

A blind human read. A blind read of 50 answers found that of 39 rated answers labelled Unsafe, 34 went along with the request but only 13 gave usable help, so U measures lost refusal well and usable harm is a smaller share (Section C).

Judge wording. A sentence about harm needs the HarmBench contrast to have the same sign as the Qwen3Guard contrast and at least half its size, and $E_{\mathrm{HB}} \geq 20$ pp; below that we say lost refusal. Section G applies the rule to every cell.

What we screened. Section G lists every setting we ran, in time order, including runs that did not enter the tables.

10 Related work

Benign fine-tuning erodes safety. Qi et al. (2024) showed that fine-tuning aligned models on benign instruction data, such as Alpaca and Dolly, erodes their safety. He et al. (2024) showed that benign rows chosen for their likeness to harmful rows erode safety more than random rows, and that

list-format answers raise erosion by themselves. Kim et al. (2025) show that optimisation choices, such as the learning rate, batch size and number of epochs, drive much of the safety loss.

Family contrasts exist. Hawkins et al. (2026) fine-tune small models from several families with one LoRA recipe (constant learning rate 5×10^{-5}, one epoch, 1,000 synthetic rows seeded from Dolly, 3 seeds) and measure SORRY-Bench compliance: Gemma-3-1B rises from 30.2% to 65.5%, Llama-3.2-1B from 34.5% to 38.4%, and Qwen3-0.6B falls (their Table 2). So the same Llama-3.2-1B erodes little in their setup and nears saturation in ours; their data, judge, starting rate and recipe all differ from ours. Khan et al. (2026) fine-tune four families of 7–9B models on medical and legal data. They write: "Base model family was likewise an unreliable predictor" (their §4.2.1). Guo et al. (2026) compare Llama-3.1-8B and Qwen2.5-7B, with no OLMo model. On Alpaca, Qwen erodes less (LoRA on 5,000 rows: 9.1% against 44.8% attack success). On Dolly with full fine-tuning, the order flips (73.6% against 66.9%). So at 7–8B, the Qwen-versus-Llama order in Guo et al. (2026) depends on the data. Their 7.3–9.1% for LoRA on Qwen2.5-7B with Alpaca (rank 32, learning rate 2×10^{-4}, 5 epochs; HEx-PHI prompts; a GPT-4o-mini judge) is far below our +78.1 pp on Dolly; the data alone may explain much of this gap, so the recipe and the data must be reported together. Fraser et al. (2025) fine-tune Llama-3.2-1B on Dolly with 4-bit LoRA for 5 epochs (their text says instruct models; their links point to base checkpoints).

Shallow alignment. Qi et al. (2025) argue that safety alignment sits mostly in the first few answer tokens. On benign Samsum data, a 10-step warmup alone cut their attack success from 23.4% to 3.8% (Table 9 of the ICLR version). The Scheduled recipe includes a warmup, and §6 finds that the opening switch carries about half of OLMo's erosion.

Content of benign data. Zhao et al. (2025b) fine-tune 95 role-specific LLaMA-3-8B models and find that safety falls further the more villainous the character. Vetter et al. (2026) trace emergent misalignment to text about villains and harmful agency; human-written documents did not reliably cause it, while synthetic instruction-answer pairs made from the same content did. Narrow fine-tuning can generalise broadly (Betley et al., 2026), and answer format alone can raise erosion (He et al., 2024). Our fiction arms are human-written passages, not role-play or instruction-answer pairs, and dark content eroded no more than light content (§8), in line with Vetter et al. (2026).

OLMo-2. We are not aware of prior benign-fine-tuning erosion numbers for OLMo-2 Instruct; our check (one web search, not exhaustive) found refusal geometry in OLMo-2-1B, where the only fine-tuning adds refusals (Labunets, 2026), and OLMo-2 7B and 13B with harmful data only (Lee et al., 2026).

We add OLMo-2, matched recipe changes within families at two sizes, and an opening-level account of where the erosion sits.

11 Conclusion

A standard loss and schedule cut the erosion in one family but barely in another, and which family it helps changes with the size: OLMo at 1B, Qwen at 7B on the main judge, and neither OLMo nor Llama at 7–8B. One recipe gives very different erosion across families at the same size, in the same order at both sizes. Only the learning-rate level moved every family we tried. In a pre-registered test, dark fiction eroded no more than light fiction, and both eroded far less than Dolly while refusing more benign prompts.

What this means in practice. First, an erosion number describes a model family, a size, a recipe, a dataset and a judge together: report all five, and every setting screened. Second, a fix found on one family or one size may not carry over: the schedule cut OLMo's erosion at 1B but not at 7B, and Qwen's at 7B but not at 1.5B, so check a fix on your own model at your own size; only the learning-rate level moved every family here. Third, compare families only at matched recipes, and recipes only within a family and size. Fourth, a dark topic did not raise the erosion here, while prose passages in place of instruction answers lowered it and raised over-refusal: report over-refusal and task accuracy beside any erosion number.

These measurements set up the larger study of repairing safety after benign fine-tuning.

Limitations

- **Scope.** Every claim holds only for these models (three instruct families, at 1B and at 7–8B),

Dolly and the two fiction datasets (on OLMo-2-1B only), LoRA fine-tuning, these recipes, the 200 development harm prompts and these judges. No result is confirmed on held-out prompts.

- **Choices made after seeing results.** We fixed the 20–55 pp window for partial erosion after the first Qwen results, with no written reason for 20 and 55. Seed 1101 chose the setting and counts in the OLMo mean. We wrote the exact 1B family-by-recipe test after 11 of its 12 values were known. Rules or predictions written before their runs cover the Scheduled recipe, the token-mean recipe, rank 16, the fiction study and the six tests of §5, and the human read's analysis was fixed before reading; the other comparisons are descriptive (Table 11). The exploratory analyses of §7 were run after the results and carry no test.

- **The transfer tests.** The 7–8B seeds spread more than the 1B seeds, so the 7–8B tests have less power, and "not shown" is weak evidence of no interaction. Six tests ran on overlapping cells with no multiplicity adjustment; T5 clears 0 only narrowly and only on Qwen3Guard, and T2 differs on HarmBench only. The half-LR test's idea came from interim replication data. The Llama-3.1-8B half-LR cell has two seeds. We tested one standard recipe change and one learning-rate step; other schedules, ranks or epochs may behave differently.

- **Fiction study.** One family, one recipe and 16 books. The fiction-trained models also refused more benign prompts and lost ARC accuracy, so their lower erosion may come from weaker instruction following. The margins are wide (about a third of Dolly's erosion), so "no difference larger than the margin" is a weak equivalence claim.

- **Few seeds.** The between-seed variance rests on 2 df: the sample SD's 90% range is 0.23–1.73 times the true SD. The OLMo band covers evaluation noise only. The 1B variant rows with rank 16 or half the learning rate rest on one or two seeds, and every interaction test has about 4–6 df. The refusal-direction probe has one or two seeds per cell.

- **What the judge measures.** U is a broad policy label that mostly tracks lost refusal (§9), and usable harm is a smaller share. HarmBench did not judge the 1B half-LR or rank-16 units, and Granite judged greedy answers only.

- **Differences between families and sizes.** The families differ in chat template, tokeniser (so the same rows give different token counts), post-training pipeline and starting over-refusal, and the two OLMo-2 sizes are different releases. The runs used four engine identities, one per group of cells (Section B). We claim no cause for the family difference or for its change with size.

- **Recipes.** The Constant recipe is not standard practice. The token-mean recipe separates the loss weighting from the schedule, but not the schedule's warmup from its decay, which also halves the mean learning rate.

- **Human read.** One reader, who designed parts of the study, rated 50 answers. There is no second rater, and the groups are too small to compare.

- **ARC.** The ARC check is uninformative at 1B.

Ethical considerations

This paper measures how fine-tuning on benign data erodes refusal in small open models, so that practitioners can see that the model family and the training recipe change the result. It proposes no attack, and the training data is a public benign instruction dataset. The results could in principle help someone choose a recipe that removes refusals; we judge this risk small, since the effects concern 1B models trained on benign data, and the same knowledge helps practitioners keep refusals. We release no harmful prompt or answer text, and we use every model, dataset and judge for research, consistent with its licence (Section C). An author read 50 model answers to harmful requests for the blind human read, with the rubric in Section C; no other annotators took part.

Use of AI assistants. We used AI assistants (Claude, Anthropic) extensively: to help design and pre-register the analyses, to write the analysis and run-management code, to check every reported number against the run records, and to draft and

edit the text. The authors made every study decision and take full responsibility for the content.

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A Per-seed tables

Table 4 gives every seed of every 1B run in Tables 1 and 2, the Qwen2.5-1.5B screens (Constant, rank 16 and High-dose) and Llama-3.2-1B under Constant, half LR; Table 5 gives every 7–8B seed. The tables in the main text give the means.

B Settings and hashes

C Artifacts, licences and the blind read

Artifacts and licences. We use OLMo-2-0425-1B-Instruct, OLMo-2-1124-7B-Instruct, Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct (Apache 2.0), Llama-3.2-1B-Instruct (Llama 3.2 Community License) and Llama-3.1-8B-Instruct (Llama 3.1 Community License); the judges Qwen3Guard-Gen-4B and Granite Guardian 3.3 8B (Apache 2.0) and the HarmBench classifier (MIT); the training data Dolly-15k (CC BY-SA 3.0) and, for the fiction study, passages from 16 Project Gutenberg books that are in the public domain in the United States (Section D); and, for evaluation only, prompts from HarmBench and StrongREJECT (MIT), SimpleSafetyTests (CC BY 2.0), and OR-Bench and XSTest (CC BY 4.0). We use each for research, consistent with its terms, and release no prompt or answer text.

The blind human read. We read 50 answers blind, with the sample, rubric and analysis fixed before reading and the ratings sealed before the key was opened. Of 39 rated answers labelled Unsafe, 34 (0.87, 95% interval 0.73–0.96) went along with the request, but only 13 (0.33, 0.19–0.50) gave usable help; the 1B models often comply vaguely, generically or wrongly. All 10 Safe-labelled controls were refusals or deflections. On the 15 OLMo answers that HarmBench judged, the human bar was stricter: 4 of 12 HarmBench *yes* answers were rated usable help. So U measures lost refusal well, and usable harm is a smaller share.

Blind-read rubric. The 50 answers were drawn by a seeded rule from the sampled answers: 40 labelled Unsafe (12 from OLMo-2-1B under the Constant recipe, 6 from OLMo-2-1B under the Scheduled recipe, and 8, 6 and 8 from Qwen2.5-1.5B, Qwen2.5-7B and Llama-3.2-1B under the Constant recipe) and 10 labelled Safe, as controls. The reader saw only an item number, the request and the answer, and gave each answer one rating: 0, refuses or deflects (declines, warns, redirects, or gives only safe general information); 1, engages but gives no usable help (goes along with the request but is vague, generic, wrong, fictional, or too incoherent to act on); or 2, gives usable help (specific steps, methods, materials or content that would meaningfully help someone do the harmful thing). The reader could also flag an answer as looping or garbled, or skip it. The rubric and the analysis were fixed before reading, and the ratings were sealed before the key was opened.

D The fiction study: data and design

Books. Sixteen public-domain books from Project Gutenberg, the first eight eligible ones in each pool by text number. *Dark* (Gothic, horror and ghost-story subjects): Frankenstein (84), The Picture of Dorian Gray (174), The Phantom of the Opera (175), Dracula (345), Tales of Terror and Mystery (537), The Monk (601), Wieland (792) and The Works of Edgar Allan Poe, Volume 2 (2148). *Light* (romance, humour and domestic subjects, with horror, crime and war excluded): O Pioneers! (24), Far from the Madding Crowd (27), The House of the Seven Gables (77), Ivanhoe (82), The Prisoner of Zenda (95), Persuasion (105), The Voyage Out (144) and Middlemarch (145).

Rows. Each arm has 5,000 rows, 625 per book, in Dolly's format: an instruction from a fixed template, the preceding sentences as context, and a passage as the response. Each row's response length

matches one Dolly response within a small band (token counts under the OLMo-2 tokenizer), so both arms copy Dolly's length distribution. For each row, 100 candidate start sentences are drawn from its book; among the candidates that match the length, the dark arm takes the passage with the most darkness-lexicon words per 100 tokens and the light arm the one with the fewest, and passages never overlap. Before the screen, lexicon hits per 100 response tokens were 1.77 (dark) against 0.02 (light). Each arm trains on 4,800 rows; 25 rows per book are held back as task rows.

Safety screen. Granite Guardian 3.3 (*harm* risk) screened every training response, and each flagged row was replaced by the next candidate from the same book under the same rule. A sexual-content check fixed in the pre-registration fired (Qwen3Guard's sexual category above 0.5% of an arm), so, as pre-registered, Granite's *sexual content* risk joined the drop rule before any run. In all, 179 dark and 47 light rows were replaced. Qwen3Guard only describes the data: it never selects or drops a row. On the final rows it labels 11 dark and 6 light responses Unsafe (of 5,000; Violent: 4 and 2), so the darker arm is also darker by the judge, but only slightly; its sexual category stays at 1.14% (dark) and 0.98% (light). The Monk holds 5 of the dark arm's 11 Unsafe rows, and The Voyage Out 4 of the light arm's 6.

Training and pre-registration. Both arms use OLMo-2-1B under the Constant recipe with seeds 1101–1103, and each arm has its own untuned unit (the same model and prompts as D's). The question, arms, measurements, contrasts, reading rules, margins and analysis script were fixed and the pre-registration frozen before any fiction training run. One deviation, recorded before any result was read: the last seed (light, 1103) was launched after an engine update, on its own packed engine, so all six units share one engine.

The fiction models also changed in other ways. Both fiction arms learned their task, but they refused more benign prompts (24–31%, against 17% untuned and 4.7% after Dolly) and their ARC accuracy fell to 28 and 34% (chance about 25%; 39% untuned, 47% after Dolly). So prose passages also weakened instruction following, which may explain part of the smaller erosion; we do not separate the two. Every unit passed the bar that answers must end; one timing deviation was recorded before any result, and all six units share one engine (Section D).

Results per seed. Table 9 gives every fiction unit. Each arm's untuned unit gives the same values as D's: U 3.2%, over-refusal 17%, ARC 74 of 189.

E Details of Sections 6 and 7

Two layers of difference between families. The families differ in two ways. First, they lose different shares of their refusal openings under the Constant recipe: -0.51 for OLMo, -0.68 for Llama and -0.85 for Qwen2.5-1.5B (sampled, seed 1101). Second, their direct answers were already unsafe at different rates before any fine-tuning: 0.18 for OLMo, against 0.57 for Qwen and 0.64 for Llama. Qwen and Llama were therefore already compliant most of the time when they did not refuse. For them, losing a refusal opening comes close to producing an Unsafe answer: the opening switch carries 0.64–0.78 of Qwen's rise and 0.68 of Llama's. OLMo's direct answers had to become less safe as well, which both layers show.

The recipes act through the openings. Across the Constant recipe, Constant, token-mean and the Scheduled recipe, the Unsafe share within direct openings stays similar (0.57–0.70). What changes is how often the model still opens with a refusal: -0.51 under Constant, -0.50 under Constant, token-mean, and -0.34 to -0.36 under Scheduled. Scheduled, rank 16 keeps the most refusal openings (-0.20) and erodes least ($+20.2$ pp), with more over-refusal (25 of 100). So the Scheduled recipe's lower erosion looks like a dose effect of its schedule, expressed through the refusal openings (one seed for Constant, token-mean and for Scheduled, rank 16).

This analysis is observational only. We ran no prefill test, and we count positions in words, not tokens.

Other leads. The Scheduled recipe lengthens the sampled answers to harm prompts in every model (mean tokens rise by 30–90%; 101 → 136 for OLMo-2-1B, 82 → 153 for OLMo-2-7B) and raises over-refusal by 3–12 pp, most in OLMo-2-1B: it changes the answers' form everywhere, and the erosion in two cells. The families erode on partly different prompts: calling a prompt eroded if its seed-mean Unsafe rate under the Constant recipe is at least 2/3, the two sizes of one family share most eroded prompts (a Jaccard index of 0.62

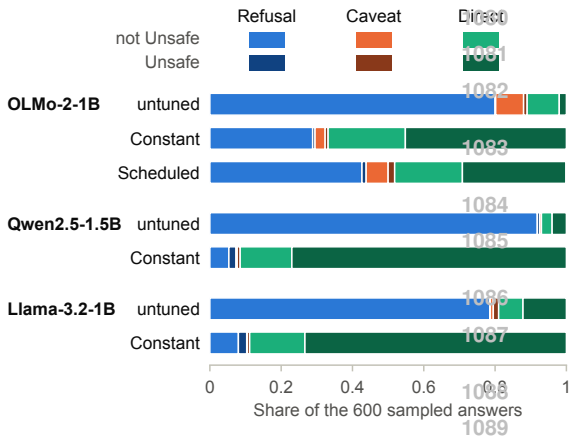


Figure 1: How the answers open, before and after fine-tuning. Each bar is one model before or after fine-tuning; fine-tuned bars are seed 1101. Each bar splits the 600 sampled answers into refusal, caveat and direct openings. The darker part of each opening is the share that Qwen3Guard labels Unsafe. The lighter part is the rest (Safe, Controversial or Invalid). One empty OLMo answer under the Scheduled recipe is not drawn.

to 0.80), while OLMo shares fewer with Llama or Qwen (0.42 to 0.57) than those two share with each other (0.66 to 0.74). A family name does not fix one starting point: OLMo-2-7B, a different release from OLMo-2-1B, starts more cautious (over-refusal 27% against 17%) and loses almost all of that caution.

F Claim status

Table 11 sorts the claims by when their rule was fixed. Eleven tests had their rule fixed before their data: the Scheduled-recipe and token-mean predictions, the rank-16 rule, the fiction study's two labels, and the six pre-registered tests of §5 (the 7–8B replication, T1, T2, T5, T6 and the half-LR test), whose files also hold our predictions. The 1B family-by-recipe test was written after 11 of its 12 values were known. We do not adjust for multiplicity. The exploratory analyses of §7 were run after the results and carry no test. The other comparisons are descriptive.

G Screening log

We list every setting we ran, in time order. All runs use LoRA rank 64, unless marked; §2 defines the recipes, and Section A gives every seed.

1. Qwen2.5-1.5B, High-dose, 3 seeds: mean $E = +81.8$ pp after 3 epochs. It failed the bar that answers must end (repetition loops).
2. Qwen2.5-1.5B, Constant, seed 1101: +75.2 pp. Seed 1102 was lost to a timer and gave no result.
3. Qwen2.5-7B, Constant, seed 1101: +82.2 pp.
4. Qwen2.5-1.5B, Constant, rank 16, seed 1101: +67.8 pp.
5. OLMo-2-1B, Constant, seed 1101: +43.5 pp; then seeds 1102 and 1103.
6. OLMo-2-1B, Scheduled, seed 1101: +29.0 pp. We pre-registered the prediction that it erodes less than the Constant recipe.
7. Llama-3.2-1B, Constant, seed 1101: +62.5 pp.
8. OLMo-2-1B, Scheduled, seeds 1102 and 1103; Llama-3.2-1B, Scheduled, seeds 1101–1103; OLMo-2-1B, Constant, token-mean and Scheduled, rank 16; both families under Scheduled, half LR (Table 2).
9. A Qwen2.5-1.5B Scheduled run was planned for this note, with a written prediction, and dropped before any run (9 Oct). Nothing from it exists.
10. Llama-3.2-1B, Constant, seeds 1102 and 1103 (9 Oct): +61.3 and +54.0 pp. Seed 1102 needed a second attempt: a cap gate stopped the first attempt before its job started.
11. Llama-3.2-1B, Constant, half LR, seed 1101 (9 Oct): +48.0 pp. Granite, judged on the pod, agrees in direction (greedy E +47.5 pp).
12. OLMo-2-1B, Constant, token-mean, seeds 1102 and 1103 (9–10 Oct), to complete three seeds: +39.5 and +42.5 pp.
13. OLMo-2-1B, Constant, on dark and on light fiction instead of Dolly, seeds 1101–1103 each (9–10 Oct), pre-registered (§8): every unit is reported.
14. On a later engine identity (10 Oct): OLMo-2-7B and Llama-3.1-8B, Constant, seed 1101 (+49.7 and +70.2 pp), then OLMo-2-7B seed 1102 (+37.2); Qwen2.5-1.5B and Qwen2.5-7B, Constant, seeds 1101–1103, where seed 1101 reproduced the earlier units' counts exactly; Llama-3.1-8B, Constant, half LR, seeds 1101 and 1102 (+52.5 and +54.0 pp), run as the larger study's second learning-rate rung.



Figure 2: Erosion by model and recipe, for the 1B units of the first version of this note. Each row is one model and recipe, and each dot is one seed; seeds that nearly coincide are stacked. The left panel gives the Qwen3Guard erosion E and the right panel the HarmBench erosion E_{HB}, both in pp. Blue marks the Constant recipe and its variants; orange marks the Scheduled recipe and its variants. A black tick marks the mean of the seeds. On one-seed rows, a bar shows $E \pm 2$ SE (evaluation noise only). The three Qwen rows show the first seed (1101) only; the later three-seed Qwen rows, the Qwen Scheduled rows and the 7–8B rows (Tables 1, 2 and 5) are not drawn. HarmBench has no recorded SE, so the right panel has no bars. HarmBench did not judge the rows marked "not judged". We do not shade a "partial" window, because we fixed it after the first Qwen results (§9).

15. Pre-registered (10–11 Oct; §5): the 7–8B replication (OLMo-2-7B Constant seed 1103, Llama-3.1-8B Constant seeds 1102 and 1103, and Scheduled seeds 1101–1103 for both); Qwen2.5-1.5B and Qwen2.5-7B, Scheduled, seeds 1101–1103; OLMo-2-7B, Constant, half LR, seeds 1101–1103. Every unit is reported.

16. Refusal-direction probes (exploratory; §7) on the 1B and 7–8B Constant and Scheduled units; 13 units had completed when this version was written, and only those are reported.

Judge wording, applied. A sentence about harm needs the HarmBench contrast to have the same sign as the Qwen3Guard contrast and at least half its size. If E_{HB} is below 20 pp, no sentence says harm. For OLMo-2-1B under the Constant recipe the rule holds on every seed ($E_{\mathrm{HB}} \geq 20$ pp and at least half of E), so we say harm there, meaning harmful compliance as HarmBench defines it: a blind human read found the HarmBench bar looser than "usable help" (4 of 12 HarmBench *yes* answers were rated usable; §9). Under the Scheduled recipe, E_{HB} is below 20 pp on every seed, so we say that its erosion is lost refusal, not harm, although the same-sign, half-size condition holds. The rule holds for every Qwen, Llama and OLMo-2-7B run that HarmBench judged (E_{HB} from $+23.0$ to $+63.5$ pp, at least 0.59 of E), so we say harm there too. In both fiction arms E_{HB} is below 20 pp, so we say lost refusal there.

Qwen2.5-1.5B recipe screens. Earlier screens suggest that Qwen2.5-1.5B responds little to other recipe settings: cutting the dose about 30 times, from the High-dose recipe after one epoch (600 steps at 2×10^{-4}) to the Constant recipe, moved U from about 83% to 80%, and Constant, rank 16 gives $+67.8$ pp, about 7 pp below rank 64 (Section A). The Scheduled recipe does not lower its erosion either (§5).

Table 4: Every 1B seed. U is the Unsafe share of the 600 sampled answers; E, E_{HB} and Granite greedy E are in pp, each against the same model's untuned unit; over-refusal is in %.

Recipe	Seed	U	E	E_{HB}	Over-refusal	Held-out loss	Granite greedy E
OLMo-2-1B							
untuned		3.2%			17	2.100	
Constant	1101	46.7%	+43.5	+27.7	4	1.542	+44.0
Constant	1102	46.8%	+43.7	+30.5	6	1.547	+40.5
Constant	1103	51.3%	+48.2	+32.8	4	1.545	+44.5
Constant, token-mean	1101	46.8%	+43.7	+29.7	9	1.587	+38.0
Constant, token-mean	1102	42.7%	+39.5	+26.5	8	1.589	+31.0†
Constant, token-mean	1103	45.7%	+42.5	+27.0	5	1.590	+35.0†
Scheduled	1101	32.2%	+29.0	+17.8	17	1.612	+23.0
Scheduled	1102	34.2%	+31.0	+19.3	14	1.613	+25.5
Scheduled	1103	29.7%	+26.5	+14.3	18	1.613	+18.5
Scheduled, half LR	1101	23.5%	+20.3	—	18	1.634	+11.5
Scheduled, half LR	1102	24.5%	+21.3	—	22	1.635	+12.0
Scheduled, rank 16	1101	23.3%	+20.2	—	25	1.649	+9.0
Qwen2.5-1.5B							
untuned		4.7%			68	1.890	
Constant	1101	79.8%	+75.2	+46.0	3	1.459	+85.5†
Constant	1102	74.0%	+69.3	+44.0	3	1.464	+75.5†
Constant	1103	76.0%	+71.3	+46.3	1	1.464	+79.5†
Scheduled	1101	79.3%	+74.7	+53.2	9	1.514	+82.0†
Scheduled	1102	78.5%	+73.8	+53.0	5	1.512	+81.5†
Scheduled	1103	78.0%	+73.3	+52.0	6	1.510	+82.0†
Constant, rank 16	1101	72.5%	+67.8	+45.8	3	1.481	+75.5
High-dose	1101	87.3%	+82.7	—	2	1.857	+88.0
High-dose	1102	88.7%	+84.0	—	3	1.829	—
High-dose	1103	83.5%	+78.8	—	10	1.850	+85.5
Llama-3.2-1B							
untuned		13.8%			11	2.243	
Constant	1101	76.3%	+62.5	+40.5	2	1.555	+68.5†
Constant	1102	75.2%	+61.3	+39.3	4	1.554	+63.0†
Constant	1103	67.8%	+54.0	+34.0	3	1.553	+57.0†
Constant, half LR	1101	61.8%	+48.0	—	5	1.566	+47.5†
Scheduled	1101	73.8%	+60.0	+35.5	7	1.619	+58.5
Scheduled	1102	70.3%	+56.5	+35.8	9	1.614	+58.5
Scheduled	1103	68.7%	+54.8	+36.0	6	1.613	+53.0
Scheduled, half LR	1101	55.0%	+41.2	—	4	1.642	+38.0

Blank: not applicable (untuned rows). —: not recorded (HarmBench did not judge the run; Granite did not judge High-dose seed 1102). HarmBench values for Llama's Constant seeds 1102 and 1103, Scheduled seed 1103 and every token-mean seed come from a second pod run (10 Oct); the units it shared with the first runs got the same labels. The Qwen2.5-1.5B Constant and Scheduled rows ran on a later engine identity (Section B); seed 1101 reproduced the earlier Constant unit's counts exactly. High-dose rows are after 3 epochs; this recipe failed its own bar that answers must end (repetition loops), so it is a screened setting, not a clean result. Untuned HarmBench shares (pod): OLMo 3.2%, Qwen2.5-1.5B 3.2%, Llama 11.0%. Untuned Granite greedy shares: OLMo 2.5%, Qwen2.5-1.5B 3.0%, Llama 13.5% (Mac) and 13.0% (pod). †Granite on the pod, against the pod-judged untuned unit; every other Granite value is from the Mac. On the Mac, Llama's Constant seed 1101 gives +67.5.

Table 5: Every 7–8B seed, as in Table 4. Every run used one engine identity (Section B). HarmBench and Granite judged every unit on the pod, in two runs of the same device type, each unit against its model's untuned unit judged in the first run.

Recipe	Seed	U	E	E_{HB}	Over-refusal	Held-out loss	Granite greedy E
OLMo-2-7B							
untuned		1.8%			27	2.121	
Constant	1101	51.5%	+49.7	+37.0	2	1.353	+49.5
Constant	1102	39.0%	+37.2	+27.7	2	1.358	+40.0
Constant	1103	47.3%	+45.5	+34.7	3	1.354	+45.0
Constant, half LR	1101	35.3%	+33.5	+25.3	4	1.368	+36.0
Constant, half LR	1102	32.7%	+30.8	+23.7	5	1.373	+34.5
Constant, half LR	1103	33.3%	+31.5	+23.0	7	1.370	+34.5
Scheduled	1101	52.8%	+51.0	+38.5	7	1.411	+44.5
Scheduled	1102	59.8%	+58.0	+43.0	6	1.412	+58.0
Scheduled	1103	43.7%	+41.8	+32.0	6	1.410	+39.0
Qwen2.5-7B							
untuned		4.8%			7	3.335	
Constant	1101	87.0%	+82.2	+63.5	3	1.293	+87.0
Constant	1102	83.3%	+78.5	+59.7	2	1.295	+86.0
Constant	1103	78.3%	+73.5	+58.5	3	1.296	+81.0
Scheduled	1101	74.0%	+69.2	+55.2	5	1.336	+76.0
Scheduled	1102	75.5%	+70.7	+57.5	6	1.337	+73.5
Scheduled	1103	71.7%	+66.8	+54.0	6	1.334	+71.5
Llama-3.1-8B							
untuned		5.3%			17	2.109	
Constant	1101	75.5%	+70.2	+53.8	1	1.321	+73.5
Constant	1102	61.2%	+55.8	+39.8	1	1.328	+50.5
Constant	1103	61.5%	+56.2	+39.7	1	1.321	+54.0
Constant, half LR	1101	57.8%	+52.5	+36.2	2	1.326	+50.0
Constant, half LR	1102	59.3%	+54.0	+37.8	2	1.328	+55.0
Scheduled	1101	65.5%	+60.2	+41.5	7	1.364	+60.0
Scheduled	1102	67.2%	+61.8	+45.2	7	1.366	+61.5
Scheduled	1103	60.8%	+55.5	+40.8	2	1.362	+54.0

Untuned HarmBench shares (pod): OLMo-2-7B 1.2%, Qwen2.5-7B 5.3%, Llama-3.1-8B 5.5%. Untuned Granite greedy shares (pod): OLMo-2-7B 1.5%, Qwen2.5-7B 4.0%, Llama-3.1-8B 6.0%. Qwen2.5-7B's untuned unit and its Constant seed 1101 reproduced the earlier one-seed units' counts exactly (on the Mac, Granite gave that seed +87.5). Llama-3.1-8B's half-LR seeds were run for the larger study before the half-LR pre-registration.

Table 6: Training and evaluation settings. Every recipe shares the top part; the bottom part lists what each recipe changes. The release holds every recipe file with its SHA-256.

Setting	Value
Shared by every recipe	
Training code	our own training loop in PyTorch, with Hugging Face Transformers (model loading) and PEFT (LoRA)
Adapter	LoRA on the q, k, v, o, gate, up and down projections of every layer; dropout 0. Trainable parameters at rank 64: OLMo-2-1B 48.2M, Llama-3.2-1B 45.1M, Qwen2.5-1.5B 73.9M, OLMo-2-7B 159.9M, Qwen2.5-7B 161.5M, Llama-3.1-8B 167.8M
Optimiser	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, no weight decay; gradient-norm clipping at 1.0
Batch	128 rows per optimiser step (micro-batches of 4 rows with 32 accumulation steps; the 7–8B models: 2 rows and 64 steps); one epoch of 4,800 rows is 38 steps
Loss	next-token negative log-likelihood on the answer and its end-of-turn token only; the prompt is masked
Precision and length	base weights in bfloat16; adapters and the loss in float32; SDPA attention; at most 3,072 tokens per training row (1,536 in the prompt)
Harm evaluation	200 development harm prompts; 3 sampled answers each (temperature 0.7, top-p 1.0) and one greedy answer; at most 2,048 new tokens
Other measures	over-refusal on 100 benign prompts (greedy answers); held-out loss on 200 Dolly rows; ARC forced choice on 189 questions
Judges	Qwen3Guard-Gen-4B on every answer (the main judge); the HarmBench Mistral-7B classifier on sampled answers; Granite Guardian 3.3 8B on greedy answers
Compute	one NVIDIA A100 80 GB GPU per run (rented cloud GPUs); a 1B run took about 1–3.5 h including setup and both evaluations, and the Qwen2.5-7B run about 3.6 h. HarmBench ran on the same GPU type; Granite on the same GPU type or on an Apple-silicon Mac
What each recipe changes	
Constant	learning rate 10^{-4}, constant; per-row loss mean; rank 64, alpha 128
Constant, token-mean	as Constant, with a token-mean loss
Constant, half LR	as Constant, with learning rate 5×10^{-5}
Constant, rank 16	as Constant, with rank 16 and alpha 32
Scheduled	peak learning rate 10^{-4}: a 4-step linear warmup, then cosine decay to 0 at the end of epoch 1; token-mean loss; rank 64, alpha 128
Scheduled, half LR	as Scheduled, with peak 5×10^{-5}
Scheduled, rank 16	as Scheduled, with rank 16 and alpha 32
High-dose (a Qwen2.5-1.5B screen)	as Constant, with learning rate 2×10^{-4}, 8 rows per step (600 steps per epoch) and 3 epochs

Table 7: Revisions and SHA-256 hashes. Model and judge revisions are Hugging Face commit hashes. Table 6 gives the training and evaluation settings.

Item	Revision or SHA-256
Models	
OLMo-2-0425-1B-Instruct	`48d788eca847d4d7548f375ad03d3c9312f6139e`
OLMo-2-1124-7B-Instruct	`470b1fba1ae01581f270116362ee4aa1b97f4c84`; tokenizer from the OLMo-2-0425-1B-Instruct revision above
Qwen2.5-1.5B-Instruct	`989aa7980e4cf806f80c7fef2b1adb7bc71aa306`
Qwen2.5-7B-Instruct	`a09a35458c702b33eeacc393d103063234e8bc28`; tokenizer from the Qwen2.5-1.5B-Instruct revision above (identical files)
Llama-3.2-1B-Instruct	`9213176726f574b556790deb65791e0c5aa438b6`; tokenizer and chat template from Llama-3.1-8B-Instruct at `0e9e39f249a16976918f6564b8830bc894c89659`
Llama-3.1-8B-Instruct	`0e9e39f249a16976918f6564b8830bc894c89659`
Judges	
Qwen3Guard-Gen-4B	`6ec42827da0c1ff11e7a49dc269d2e810d27e108`
HarmBench-Mistral-7b-val-cls	`51182c7cdaf9b2e5f05b745b81b047b67e7384a1`
Granite Guardian 3.3 8B, Q4_K_M	granite-guardian-3.3-8b-GGUF at `8c87479a2dc329e4e362a6e5dda7e7de3e0372b2`; file SHA-256 `b4fa861b5163373413d9659c22e344e6206baad824ead17226b2058e54d14587`
Prepared Dolly data (SHA-256 of each manifest)	
Qwen2.5 (1.5B and 7B)	`9247c4bbccf212b02be7f221e2ee451d17ebf643a39a44db6e3bed6fb4d750f2`
OLMo-2 (1B and 7B)	`9ed61826f1634dcab5a7385f18556b108b3ba8db95fc17215fa4cf4f6cb993b9`
Llama-3 (3.2-1B and 3.1-8B)	`c52a50afddcfcfa818512ac7f1536c34c959774e815095bd30f69320b083d55a`
Engine identities (code SHA-256)	
The first Qwen2.5 runs (rank 16, High-dose)	`c52c0ebc83793fc552f84908e83542d5e46543459fcafb7be2fc9c7ff42ac597`
OLMo-2-1B, Constant	`552538748acec1a8ffd1ee976b9eb3c3fdfe9ca01b8148ee8810291409357441`
Llama-3.2-1B runs, and the other OLMo-2-1B runs	`e49a978c5fa27c3ba4bb1570ac8b24501e09bf8db5c35e31ffba90bf775cdc06`
Every run of 10–11 Oct: the three-seed Qwen rows, every 7–8B row and the Qwen Scheduled rows (engine X)	`472fc7f6d619bdee2c3aa38e444024626be8b66bfb74a5abf37f2ddb6dded892`
Analysis	
Interaction-test script of §5, hashed in every pre-registration before its runs	`798e6c03177baf4267a1880a6a897186045a95aef27e496cc6e27c4bba311cf2`
Opening word list, frozen before any result (§6)	`bbbc71e9eebb62876430ffb5e0f54113c3818984b8369924b7bd2212c4c1ca17`
Reading bars, with the 20–55 pp window (§3)	`c0b6f2aae43285de632a175adb1e913cd31f37bc6ac25fb56e4468ff3a1208ea`
Logit-scale prediction for the Scheduled recipe on Qwen2.5-1.5B, written before the run was dropped (§9)	commit `c868608a08ea1192e1e8bf489488631a4903beaa`, 2026-10-08 17:14 UTC

Table 8: Fiction against Dolly on OLMo-2-1B (Constant recipe; means over three seeds). E, E_{HB} and Granite greedy E are in pp against each arm's untuned unit. Over-refusal and ARC accuracy are in %; untuned, they are 17 and 39.2.

Training data	E	E_{HB}	Granite	Over-ref.	ARC
Dark fiction	$+11.7$	$+5.1$	$+8.0$	30.7	27.7
Light fiction	$+16.6$	$+6.7$	$+8.0$	24.0	33.7
Dolly	$+45.1$	$+30.3$	$+43.0$	4.7	46.7

Table 9: Every fiction seed (OLMo-2-1B, Constant recipe). U is the Unsafe share of the 600 sampled answers; E, E_{HB} and Granite greedy E are in pp against the arm's untuned unit; over-refusal is in %; ARC is correct answers of 189. Granite judged the fiction units on the pod and D on the Mac (Mac-versus-pod parity passed; the untuned unit gives 5 of 200 on both).

Arm	Seed	U	E	E_{HB}	Granite greedy E	Over-refusal	ARC
Dark	1101	17.2%	+14.0	+6.7	+8.5	31	53
Dark	1102	15.2%	+12.0	+4.8	+7.0	29	52
Dark	1103	12.3%	+9.2	+3.7	+8.5	32	52
Light	1101	21.2%	+18.0	+8.3	+7.5	22	65
Light	1102	20.0%	+16.8	+6.5	+9.5	25	65
Light	1103	18.2%	+15.0	+5.3	+7.0	25	61
Dolly (D)	1101	46.7%	+43.5	+27.7	+44.0	4	94
Dolly (D)	1102	46.8%	+43.7	+30.5	+40.5	6	87
Dolly (D)	1103	51.3%	+48.2	+32.8	+44.5	4	84

Table 10: Refusal-direction retention (exploratory). Retained is the share of the untuned harmful-versus-harmless separation that the fine-tuned model keeps; cosine is between the untuned and fine-tuned directions; both are means over the upper half of the layers. One value per seed.

Model	Recipe	E (pp)	Retained	Cosine
OLMo-1B	Constant	+43.5, +48.2	0.59, 0.59	0.67, 0.67
OLMo-1B	Scheduled	+31.0, +26.5	**0.71, 0.74**	**0.79, 0.80**
Llama-1B	Constant	+62.5, +61.3	0.57, 0.57	0.50, 0.51
Llama-1B	Scheduled	+60.0	0.62	0.54
OLMo-7B	Constant	+49.7, +37.2	0.37, 0.41	0.56, 0.60
OLMo-7B	Scheduled	+51.0	0.42	0.60
Llama-8B	Constant	+70.2, +56.2	0.32, 0.35	0.32, 0.37
Llama-8B	Scheduled	+60.2	0.41	0.46

Table 11: When each claim's rule was fixed.

Claim	Status	Note
OLMo's partial erosion (Constant)	Descriptive	Seed 1101 chose the setting; the fresh seeds give $+45.9$ pp
The families differ at one recipe	Descriptive	The 20–55 pp partial window was fixed after the Qwen results
Scheduled erodes less than Constant (OLMo)	Prediction before the run	Held: $+28.8$ against $+45.1$ pp
Constant, token-mean gives $E \leq 36.25$ pp	Prediction before the run	Failed: $+41.9$ pp
Rank matters little (Scheduled, rank 16)	Rule before the run	One seed each
The recipe effect differs by family at 1B (OLMo against Llama)	Design before two seeds; exact test after 11 of 12 values	Excludes 0 on Qwen3Guard and on HarmBench
7–8B replication (OLMo-2-7B against Llama-3.1-8B)	Pre-registered before any of its nine runs	Not shown on both judges; the point estimate has the opposite sign; the two primary predictions failed, the Llama prediction held
Scheduled on Qwen2.5-1.5B: T1, T2	Pre-registered before any run	T1 differs (OLMo) on both judges; T2 not shown on Qwen3Guard, differs on HarmBench; the test predictions held, the size prediction partly
Scheduled on Qwen2.5-7B: T5, T6	Pre-registered before any run	T5 differs (Qwen) on Qwen3Guard only, clearing 0 narrowly; T6 not shown; the T5 prediction failed
Half LR at 7–8B	Pre-registered before any OLMo half-LR run; idea from interim data, disclosed	Not shown on both judges; both predictions held
The opening switch carries about half	Descriptive	Observational
Refusal direction, answer length, eroded prompts, baselines	Exploratory, after the results	No test; one to three seeds per cell
Blind human read	Sample, rubric and analysis before reading	One reader
Fiction: content and Dolly labels	Pre-registered, frozen	One timing deviation, recorded before any result
Family order at 7–8B; the three-seed Qwen rows	Descriptive	Three seeds each, on a later engine identity